Funkcje PostgreSQL

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CASE

	last_name	salary	warunek
1	King	24,000	OK
2	Yang	17,000	OK
3	Garcia	17,000	OK
4	James	9,000	---
5	Miller	6,000	---
6	Williams	4,800	---
7	Jackson	4,800	---
8	Nguyen	4,200	---
9	Gruenberg	12,008	OK

```sql
SELECT
    last_name,
    salary,
    CASE
        WHEN salary > 10000 THEN 'OK'
    ELSE '---'
    END AS warunek
FROM
        employees
```

LOWER, UPPER, INITCAP

	last_name	lower	upper	job_id	ic
1	King	king	KING	AD_PRES	Ad_Pres
2	Yang	yang	YANG	AD_VP	Ad_Vp
3	Garcia	garcia	GARCIA	AD_VP	Ad_Vp
4	James	james	JAMES	IT_PROG	It_Prog
5	Miller	miller	MILLER	IT_PROG	It_Prog
6	Williams	williams	WILLIAMS	IT_PROG	It_Prog

```
SELECT
    last_name,
    lower(last_name) lower,
    upper(last_name) upper,
    job_id,
    InitCap(job_id) ic
FROM
        employees
```

SUBSTRING

	last_name	substr	substr2
1	King	Ki	ing
2	Yang	Ya	ang
3	Garcia	Ga	arc
4	James	Ja	ame
5	Miller	Mi	ill
6	Williams	Wi	ill

```
SELECT
    last_name,
    substring(last_name,1,2)  substr,
    substring(last_name,2,3)  substr2
FROM
    employees
```

REPLACE / TRANSLATE

	last_name	replace1	replace2	translate1
1	King	KXYZg	King	KXYg
2	Yang	Yang	YWWWng	YaYg
3	Garcia	Garcia	GWWWrciWWW	GarcXa
4	James	James	JWWWmes	James
5	Miller	Miller	Miller	MXller
6	Williams	Williams	WilliWWWms	WXllXams

```
SELECT
    last_name,
    replace(last_name, 'in','XYZ') replace1,
    replace(last_name, 'a','WWW') replace2,
    translate(last_name, 'in', 'XY') translate1
FROM
        employees
```

REVERSE, RIGHT, LEFT

	last_name	reverse	r	l
1	King	gniK	ng	Ki
2	Yang	gnaY	ng	Ya
3	Garcia	aicraG	ia	Ga
4	James	semaJ	es	Ja
5	Miller	relliM	er	Mi
6	Williams	smailliW	ms	Wi

```
SELECT
    last_name,
    REVERSE(last_name) reverse,
    RIGHT(last_name, 2) r,
    LEFT(last_name, 2) l
FROM
        employees
```

TRIM

	A-Z ze_spacja	A-Z t
1	spacja	spacja

```
SELECT
    'spacja ' ze_spacja,
    trim('spacja ') t
```

LTRIM, RTRIM, RPAD, LPAD

	last_name	rt	lt	lpad	rpad
1	Abel	Abe	bel	XXXXXXAbel	AbelXXXXXX
2	Ande	Ande	nde	XXXXXXAnde	AndeXXXXXX
3	Atkinson	Atkinson	tkinson	XXAtkinson	AtkinsonXX
4	Baida	Baida	Baida	XXXXXBaida	BaidaXXXXX
5	Banda	Banda	Banda	XXXXXBanda	BandaXXXXX

```
SELECT
    last_name ,
    rtrim(last_name,'l') rt,
    ltrim(last_name,'A') lt,
    lpad(last_name,10,'X'),
    rpad(last_name,10,'X')
FROM employees e
order by last_name
```

POSITION

	last_name	position	position
1	King	0	4
2	Yang	2	4
3	Garcia	2	0
4	James	2	0
5	Miller	0	0
6	Williams	6	0

```sql
select
    last_name,
    position('a' in last_name),
    position('g' in last_name)
FROM employees
```

LENGTH

	last_name	length
1	King	4
2	Yang	4
3	Garcia	6
4	James	5
5	Miller	6
6	Williams	8

```
SELECT
    last_name,
    length(last_name)  length
FROM
        employees
```

KONKATENACJA

	first_name	last_name	konkatenacja1	konkatenacja2
1	Steven	King	Steven King	Steven King
2	Neena	Yang	Neena Yang	Neena Yang
3	Lex	Garcia	Lex Garcia	Lex Garcia
4	Alexander	James	Alexander James	Alexander James
5	Bruce	Miller	Bruce Miller	Bruce Miller
6	David	Williams	David Williams	David Williams

```
select
    first_name ,
    last_name ,
    concat(first_name, ' ', last_name) konkatenacja1,
    first_name || ' ' || last_name konkatenacja2
from employees e
```

LICZBOWE

	round	floor	ceil	modulo	potega	pierwiastek
1	3.12	3	4	3	64	9

```
SELECT
    round(3.1234, 2) round,
    floor(3.1234) floor,
    ceiling(3.1234) ceil,
    10%7 modulo,
    power(8, 2) potega,
    sqrt(81) pierwiastek
```

DATE_PART

	hire_date	123 rok	123 miesiac	123 dzien
1	2013-06-17	2,013	6	17
2	2015-09-21	2,015	9	21
3	2011-01-13	2,011	1	13
4	2016-01-03	2,016	1	3
5	2017-05-21	2,017	5	21
6	2015-06-25	2,015	6	25
7	2016-02-05	2,016	2	5

```
SELECT
    hire_date,
    DATE_PART('year',hire_date) rok,
    DATE_PART('month',hire_date) miesiac,
    DATE_PART('day',hire_date) dzien
from employees
```

AGE

	hire_date	age
1	2013-06-17	11 years 10 mons 7 days
2	2015-09-21	9 years 7 mons 3 days
3	2011-01-13	14 years 3 mons 11 days
4	2016-01-03	9 years 3 mons 21 days
5	2017-05-21	7 years 11 mons 3 days
6	2015-06-25	9 years 9 mons 29 days

```
SELECT
    hire_date,
    age(hire_date)
from employees
```

DATE_PART

	hire_date	now	liczba_lat	liczba_miesiecy	liczba_dni
1	2013-06-17	2025-04-24 11:59:40.909 +0200	11	142	4,329
2	2015-09-21	2025-04-24 11:59:40.909 +0200	9	115	3,503
3	2011-01-13	2025-04-24 11:59:40.909 +0200	14	171	5,215
4	2016-01-03	2025-04-24 11:59:40.909 +0200	9	111	3,399
5	2017-05-21	2025-04-24 11:59:40.909 +0200	7	95	2,895
6	2015-06-25	2025-04-24 11:59:40.909 +0200	9	117	3,591

```
SELECT
    hire_date,
    now(),
    DATE_PART('year', AGE(now(), hire_date)) liczba_lat,
    DATE_PART('year', AGE(now(), hire_date)) * 12 + DATE_PART('month',
        AGE(now(), hire_date)) liczba_miesiecy,
    now()::date - hire_date::date liczba_dni
from employees
```

INTERVAL

	hire_date	dodany_rok	dodany_miesiac	dodany_dzien
1	2013-06-17	2014-06-17 00:00:00.000	2013-07-17 00:00:00.000	2013-06-18 00:00:00.000
2	2015-09-21	2016-09-21 00:00:00.000	2015-10-21 00:00:00.000	2015-09-22 00:00:00.000
3	2011-01-13	2012-01-13 00:00:00.000	2011-02-13 00:00:00.000	2011-01-14 00:00:00.000
4	2016-01-03	2017-01-03 00:00:00.000	2016-02-03 00:00:00.000	2016-01-04 00:00:00.000
5	2017-05-21	2018-05-21 00:00:00.000	2017-06-21 00:00:00.000	2017-05-22 00:00:00.000
6	2015-06-25	2016-06-25 00:00:00.000	2015-07-25 00:00:00.000	2015-06-26 00:00:00.000
7	2016-02-05	2017-02-05 00:00:00.000	2016-03-05 00:00:00.000	2016-02-06 00:00:00.000
8	2017-02-07	2018-02-07 00:00:00.000	2017-03-07 00:00:00.000	2017-02-08 00:00:00.000

```
SELECT
    hire_date,
    hire_date + interval '1 year' dodany_rok,
    hire_date + interval '1 month' dodany_miesiac,
    hire_date + interval '1 day' dodany_dzien
from employees
```

DATE FORMAT

	hire_date	format1	format2
1	2013-06-17	17-June -2013	Monday , 17-06-13
2	2015-09-21	21-September-2015	Monday , 21-09-15
3	2011-01-13	13-January -2011	Thursday , 13-01-11
4	2016-01-03	03-January -2016	Sunday , 03-01-16
5	2017-05-21	21-May -2017	Sunday , 21-05-17
6	2015-06-25	25-June -2015	Thursday , 25-06-15

```
select
    hire_date,
    to_char(hire_date, 'DD-Month-YYYY') format1,
    to_char(hire_date, 'Day, DD-MM-YY') format2
from employees e
```

https://www.postgresql.org/docs/15/functions-formatting.html

Konwersja

	?column?	123 ?column?
1	2021-05-31	2

```
SELECT
    '2021-05-30'::date + 1,
    '1'::integer+ '1'::integer
```